
LATTICE-BASED LINKABLE RING SIGNATURES FOR ANONYMOUS AND ACCOUNTABLE WHISTLEBLOWING

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ABSTRACT

Ring signatures allow an individual to sign a message on behalf of a group in such a way that the verifier can only confirm that someone in the group signed it, but cannot identify the actual signer. This strong anonymity, while desirable, may also be exploited for repeated or harmful activities. Linkable ring signatures mitigate this issue by enabling the system to recognise whether two signatures originate from the same signer, while still keeping the signer anonymous. Such constructions are essential in domains like e-voting, e-cash, privacy-preserving blockchain systems, and whistleblowing, where detecting repeated actions—such as double-spending or double-voting—is necessary to maintain system reliability.

In this paper, we present a lattice-based linkable ring signature scheme designed to withstand quantum-era adversaries. The framework relies on exact and efficient zero-knowledge proofs, and employs a weak pseudorandom function (wPRF) to enable linkability. To demonstrate both ring membership and the generation of a unique tag, we integrate a Merkle tree accumulator, which also streamlines the verification steps. The scheme is instantiated using concrete parameter choices, allowing us to precisely evaluate how the signature size scales with different ring sizes. An important feature of our design is that it eliminates the need for trapdoor techniques, yet still produces a signature of roughly 0.22 MB when the ring contains 2^{10} users. We further outline practical application scenarios, such as anonymous but accountable whistleblowing, to highlight the usefulness of the proposed construction.

1 Introduction

Ring signatures, first proposed by Rivest, Shamir, and Tauman [1], allow an individual to sign a message on behalf of a group that is formed in an ad hoc manner, while ensuring that the signer’s identity remains hidden and that no trusted authority is needed to manage the system. In contrast, group signature schemes [2] depend on a central manager responsible for issuing keys, regulating membership, and, if necessary, revealing the signer’s identity. Because ring signatures guarantee unconditional anonymity, they are particularly suited for settings such as whistleblowing, where protecting the identity of the honest reporter is essential. Despite this strong privacy, unrestricted anonymity may sometimes be exploited, creating challenges related to accountability. To mitigate such issues, Liu, Wei, and Wong introduced the concept of Linkable Ring Signatures [3]. Their model enhances classical ring signatures by incorporating a mechanism that enables the system to determine whether two signatures originate from the same, unknown signer. The linkable anonymity maintains user privacy while providing a means to detect repeated actions, thereby striking a practical balance between anonymity and accountability in sensitive applications.

The literature presents several distinct interpretations of linkability in the context of ring signatures. One classical notion is *ring-based linkability*, which permits the linking of signatures produced with the same signing key, but only when the signatures belong to the same ring. This concept appears in earlier works such as [3, 4]. A different perspective, introduced in [5], is *message-based linkability*, where signatures created with the same secret key on an identical message can be linked, even if they come from different rings. A more generalised version, termed *event-oriented linkability*, is explored in [6, 7, 8]. In this setting, each signature carries an event identifier, and two signatures are deemed related when they match both in the signer’s key and this event tag. Another frequently studied variant is *one-time linkability* [9, 10, 11], where any pair of signatures produced by the same key is linkable. Typically, the key is assumed to be used only once, ensuring that repeated signing behaviour can be detected.

Linkable Ring Signatures (LRS) play a crucial role in numerous practical settings where anonymity must be preserved without compromising the ability to detect repeated actions from the same user. They are particularly valuable in environments that require accountable anonymity, and have been adopted in applications such as electronic voting [12], electronic cash systems [13], privacy-oriented cryptocurrencies [14], ad-hoc authentication, and blockchain systems that balance privacy with auditability [15, 16]. Recent studies also highlight their usefulness in improving the reliability and confidentiality of medical peer-review processes [17]. In voting protocols, LRS helps maintain voter privacy by separating the identity of a voter from the ballot they cast, while still enabling the system to detect any attempt at submitting multiple votes. Likewise, in e-cash and cryptocurrency platforms, LRS supports private transactions but ensures that instances of double spending can be recognised and linked together. This blend of anonymity and controlled traceability makes LRS a compelling cryptographic tool for systems that must protect user privacy while preventing misuse of privileges.

Related Work: The concept of ring signatures was first introduced by Rivest et al. [1] to provide signer anonymity within a group. However, the strong anonymity offered by ring signatures can be misused in certain scenarios, such as the double-spending problem in cryptocurrencies [18], where accountability becomes necessary. To address this, various extensions of ring signatures were proposed, including linkable ring signatures (LRS) [3, 4], traceable ring signatures [5], and unique ring signatures [19]. These variants restrict the level of anonymity to enable specific forms of accountability. The introduction of LRS in [3, 4] inspired several follow-up works [20, 21, 13], but most of these constructions rely on classical hardness assumptions such as large integer factorisation and discrete logarithms, which are known to be insecure in the presence of quantum adversaries. With the growing threat of quantum computing, lattice-based cryptography is seen as a strong option for post-quantum security. This offers benefits such as efficient signature schemes, parallel processing, and simplicity. These advantages have drawn significant attention from the cryptographic community towards lattice-based ring signature schemes [22, 23, 24, 25, 26]. In 2018, Torres et al. [9] introduced the first lattice-based linkable ring signature. Since then, the field has witnessed substantial progress through a series of works [11, 27, 28, 29, 30, 31], which have focused on designing more secure, efficient, and scalable LRS constructions grounded in lattice-based hardness assumptions.

Libert et al. [22] proposed the first lattice-based ring signature scheme with a signature size that grows logarithmically with the ring size. Their construction leveraged an accumulator scheme within the zero-knowledge framework of Stern’s protocol [32]. Building upon this foundation, Yang et al. [24] introduced a lattice-based linkable ring signature by incorporating a weak pseudorandom function, while still relying on Stern-style zero-knowledge proofs. A major limitation of Stern’s protocol is its need for hundreds of repetitions to achieve negligible soundness error, which leads to large signature sizes. To address this, Yang et al. [23] presented significant improvements in the zero-knowledge proof system, resulting in more efficient and compact lattice-based ring signatures. Further advancements, such as those by Esgin et al. [33], have introduced exact zero-knowledge proofs that are both efficient and practical for real-world deployment. In this work, we aim to construct an efficient lattice-based linkable ring signature scheme along with a real-world application model that enables whistleblower anonymity while preventing the misuse of complete anonymity.

1.1 Our Results

Our construction is denoted by $\text{LRS}^+ = (\text{Setup}, \text{UKeygen}, \text{Sign}, \text{Vrfy}, \text{Link})$. Our main contributions are as follows:

- Our design utilises efficient exact zero-knowledge proofs within an appropriate setting, ensuring clarity and completeness in its description.
- We employ a Merkle tree accumulator to aggregate public keys along with a public tag, forming the basis for linkability.
- We propose a practical application scenario where our variant is particularly relevant and beneficial.
- We provide a detailed analysis of the concrete signature size, supported by well-defined parameters.

1.1.1 Contribution Overview:

Zero-Knowledge Proof Instantiation: Our scheme employs an exact zero-knowledge proof to construct a ring signature scheme with linkability, enabling accountability without compromising anonymity. Unlike the approaches that rely on trapdoors, which often lead to increased complexity. Our method eliminates this requirement through the use of zero-knowledge proofs (ZKP). For instance, the LRS scheme in [24], which adopts a Stern-like protocol [32] but has a large soundness error of $2/3$ per round, requiring hundreds of repetitions to attain negligible error. This results in an impractically large signature size. In contrast, our construction leverages exact zero-knowledge proofs, such as those proposed in [23] and [33], which achieve significantly better soundness errors of $1/\text{poly}$ and $1/q$ respectively, where q is the modulus. These improved ZKPs require only a single-digit number of repetitions, leading to a substantial reduction in signature size, thereby making our scheme far more practical for real-world applications.

Membership and Linkability: To preserve signer anonymity, Libert et al. [22] employed a Merkle tree accumulator within a zero-knowledge framework, allowing the signer to prove membership in the ring without revealing their identity. Our scheme additionally supports linkability, achieved by generating a unique tag using a weak pseudorandom function (wPRF) [24] based on the signer’s secret key. To further enhance security, this unique tag is embedded into each node of the Merkle tree, contributing to the accumulated value during the signing process. Moreover, we modify the hash function using a matrix $\tilde{C}$ to address the dimension mismatch between the unique tag and the signer’s public key, ensuring compatibility and correctness within the cryptographic structure.

Anonymous and Accountable Whistleblowing: Our signature scheme is well-suited for whistleblowing scenarios where both anonymity and accountability are crucial. Individuals hesitate to report misconduct due to fear of retaliation or damage to their reputation. Our construction enables whistleblowers to prove organizational membership without revealing their identity, while the linkability property ensures that multiple reports from the same anonymous source can be correlated to prevent abuse. To demonstrate our approach, we present a well-defined demo setup with a supporting demographic figure that illustrates the real-world applicability of the system.

Concrete Signature Size Analysis: The theoretical strength of our scheme is supported by a concrete analysis of the signature size, based on carefully chosen parameters that balance security and efficiency. We evaluate two different ZKP frameworks: Yang et al. [23] and Esgin et al. [33]. In the framework of [23], only two rounds of repetition are required to attain 80-bit, 100-bit, and 128-bit security levels, resulting in signature sizes that are comparable to or even smaller than some of the existing schemes (see Table 1). In the framework of [33], four rounds suffice to reach negligible soundness error, yielding even more compact signatures. A detailed efficiency analysis is provided in Section 6, with signature size variations illustrated against ring size in Figure 3.

Table 1: Comparison of Different Ring Signature Schemes

Scheme	Signature Growth	Linkability	Security Bits	Random Oracle	Hardness Assumption	Signature Size
[22]	Logarithm	No	128	Yes	SIVP	75.58 MB
[23]	Logarithm	No	128	Yes	SIVP	9.70 MB
[9]	Linear	Yes	100	Yes	Ring-SIS	9.77 MB
[11]	Linear	Yes	100	Yes	M-LWE, M-SIS	9.36 MB
[34]	Linear	Yes	100	Yes	NTRU	1.30 MB
Our (I)	Logarithm	Yes	80	Yes	SIS, LWR	4.70 MB
			100			6.54 MB
			128			10.73 MB
Our (II)	Logarithm	Yes	128	Yes	SIS, LWR	0.22 MB

1.1.2 Technical Overview

Our proposed signature scheme consists of five core algorithms, collectively denoted as $\text{LRS}^+ = (\text{Setup}, \text{UKeygen}, \text{Sign}, \text{Vrfy}, \text{Link})$. The Setup algorithm initialises the system by generating the public parameters required for the protocol. The UKeygen algorithm generates a user’s key pair. The Sign algorithm produces a signature on a given message and simultaneously generates a unique tag used for linkability. The Vrfy algorithm is executed by the verifier to check the validity of the signature. Finally, Link algorithm determines whether two signatures were generated by the same user, either on the same or different messages. The brief overview of the LRS^+ scheme (Section 4) is as follows:

LRS⁺ Algorithms

- 1: **Procedure:** LRS⁺.Setup(λ)
- 2: **Input:** Security parameter λ
- 3: **Output:** Public parameters

$$\text{pp} = \{n, m_1, q, p, k_1, k'_1, m_2, m', \gamma', \tilde{\mathbf{B}}, \tilde{\mathbf{C}}\}$$

- n is a polynomial in λ , $k_1 = \lceil \log q \rceil$, $k'_1 = \lceil \log p \rceil$
- $m = 2nk_1$, $m_2 \geq \frac{m_1(\log q + 1)}{\log p - 1}$, $m' = m_2k'_1$
- $\gamma' = q/p$, a positive odd integer
- $\tilde{\mathbf{B}} = [\mathbf{B}_1 \mid \mathbf{B}_2] \xleftarrow{\$} \mathbb{Z}_q^{n \times nk_1} \times \mathbb{Z}_q^{n \times nk_1}$
- $\tilde{\mathbf{T}} \xleftarrow{\$} \mathbb{Z}_q^{m_2 \times m_1}$, $\tilde{\mathbf{C}} = [\mathbf{C}_1 \mid \mathbf{C}_2] \xleftarrow{\$} \mathbb{Z}_q^{n \times m_2k'_1} \times \mathbb{Z}_q^{n \times m_2k'_1}$

- 4: **Procedure:** LRS⁺.UKeygen(pp)
- 5: **Output:** Key pair ($\text{usk}, \mathbf{p}$)
- 6: Sample $\text{usk} \xleftarrow{\$} \{0, 1\}^{m_1}$
- 7: Compute $\mathbf{p} = \text{bin}(\tilde{\mathbf{B}} \cdot \text{usk} \bmod q) \in \{0, 1\}^{nk_1}$
- 8: **Procedure:** LRS⁺.Sign(pp, usk, m, $\mathbf{S}_{pk}$)
- 9: **Input:** Message m, ring $\mathbf{S}_{pk} = \{\mathbf{p}_0, \dots, \mathbf{p}_{N-1}\}$
- 10: **Output:** Signature $\sigma = (\pi, \tilde{\mathbf{t}})$

- Generate unique tag: $\tilde{\mathbf{t}} = \text{bin}(\mathcal{F}_{\text{usk}}(\tilde{\mathbf{T}}) \bmod p)$
- Evaluate accumulator: $\mathbf{u}_{rt} \leftarrow \text{MT.Accum}(\text{pp}, \tilde{\mathbf{t}}, \mathbf{S}_{pk})$
- Generate witness:

$$\text{wit} \leftarrow \text{MT.Witns}(\text{pp}, \mathbf{p}, \tilde{\mathbf{t}}, \mathbf{S}_{pk})$$

- Construct NIZKAoK $\pi = \text{SPK}\{\dots\}[\mathbf{m}]$ such that:

$$\tilde{\mathbf{B}} \cdot \text{usk} = \mathbf{G} \cdot \mathbf{p}$$

$$\mathcal{F}_{\text{usk}}(\tilde{\mathbf{T}}) = \mathbf{G}' \cdot \tilde{\mathbf{t}}$$

$$\text{MT.Verify}(\text{pp}, \mathbf{u}_{rt}, \mathbf{p}, \tilde{\mathbf{t}}, \text{wit}) = 1$$

where:

$$\mathbf{G} = \mathbf{I}_n \otimes (1, 2, \dots, 2^{k_1-1}), \quad \mathbf{G}' = \mathbf{I}_{m_2} \otimes (1, 2, \dots, 2^{k'_1-1})$$

- 11: **Procedure:** LRS⁺.Vrfy(pp, m, $\mathbf{S}_{pk}$, σ)
- 12: **Output:** 1 (accept) or 0 (reject)
 - Compute $\mathbf{u}_{rt} \leftarrow \text{MT.Accum}(\text{pp}, \tilde{\mathbf{t}}, \mathbf{S}_{pk})$
 - Check validity of π using SPK verifier on message m with $\mathbf{u}_{rt}$ and $\tilde{\mathbf{t}}$
 - Output 1 if verification succeeds; otherwise, 0
- 13: **Procedure:** LRS⁺.Link(pp, σ_1, σ_2)
- 14: **Input:** Signatures $\sigma_1 = (\pi_1, \tilde{\mathbf{t}}_1)$ and $\sigma_2 = (\pi_2, \tilde{\mathbf{t}}_2)$
- 15: **Output:** 1 (linked) or 0 (unlinked)
- 16: **If** $\tilde{\mathbf{t}}_1 = \tilde{\mathbf{t}}_2$ **then return** 1; **else return** 0

1.2 Application: LRS⁺ for Anonymous and Accountable Whistleblowing

Real life problem:

In institutions such as government bodies, corporations, and academic organizations, individuals may become aware of unethical or illegal activities, such as fraud, corruption, or harassment. While these individuals may wish to report such misconduct (i.e., whistleblow), they often refrain due to fears of retaliation, professional consequences, or damage to personal reputation. To foster a culture of accountability and transparency, anonymous whistleblowing platforms are

increasingly adopted.

Critical challenges:

- Maintaining anonymity: The whistleblower must not be identifiable, even to the system itself.
- Preventing duplicate reports: A single whistleblower may submit multiple identical complaints. This inflates statistics and leads to biased investigations.
- Preventing identity multiplication: An attacker may pretend to be multiple whistleblowers to make a case look stronger than it is.
- Accountability without identity: If a false report is submitted, the system must be able to link repeated malicious or false submissions to the same originator, enabling action against misuse without compromising anonymity.

Solution utilizing LRS^+ :

To address the aforementioned challenges in whistleblowing systems, we propose the integration of the LRS^+ scheme. The protocol enables anonymous, linkable, and accountable reporting while preserving the reporter’s privacy. The system proceeds through the following phases:

1. **Setup Phase:** The platform initializes the system by generating the required public parameters using:

$$pp \leftarrow LRS^+.Setup(\lambda).$$

2. **Employee Key Generation:** Each employee in the organization produces a key pair by running the key generation algorithm:

$$(usk, p) \leftarrow LRS^+.UKeygen(pp).$$

3. **Anonymous Reporting:** When an employee wishes to report misconduct, they sign the message m using their private key. The generated signature $\sigma = (\pi, \tilde{t})$ includes a unique tag $\tilde{t}$ derived from wPRF. This process ensures that:

- The signer is an employee of the organisation.
- The signature is valid.
- The anonymity of the employee is maintained.

4. **Verification:** The platform verifies the signature using $LRS^+.Vrfy(pp, m, S_{pk}, \sigma)$. This confirms:

- Message integrity.
- Membership in the organisation.
- Non-repudiability.

5. **Linkability for Accountability:** To detect repeated or malicious submissions, the platform executes $LRS^+.Link(\sigma_1, \sigma_2)$:

- If $\tilde{t}_1 = \tilde{t}_2$, the signatures are linked, indicating the same reporter.
- Otherwise, the reporters are considered distinct.

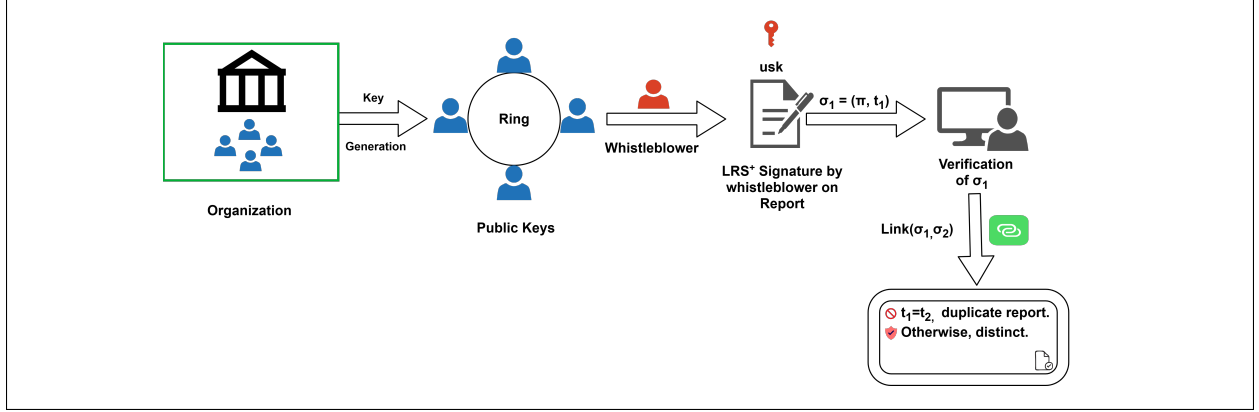
The proposed system built on LRS^+ offers a robust balance of anonymity, accountability, and integrity. The anonymity of the whistleblower is preserved through ring signatures combined with zero-knowledge proofs, ensuring the reporter cannot be identified even by the platform. Proof of membership is guaranteed using Merkle tree accumulators, with ZK proofs verifying inclusion without revealing the signer’s identity. To detect spam or duplicate reports, the system leverages tag-based linkability, enabling the identification of repeated submissions from the same source. Impersonation is prevented as valid signatures can only be generated by holders of secret keys within the authorised ring. A demonstrative overview of the system is provided in Figure 1.

2 Preliminaries

2.1 Notations

In this paper, we follow these notational conventions:

- Bold lowercase symbols (e.g., $\mathbf{u}$) indicate column vectors, whereas bold uppercase symbols (e.g., $\mathbf{A}, \tilde{\mathbf{A}}$) denote matrices.

Figure 1: Anonymous Whistleblowing Platform Using LRS⁺

- The transpose of a vector $\mathbf{u}$ is expressed as $\mathbf{u}^T$.
- Concatenating two vectors $\mathbf{u}$ and $\mathbf{v}$ is represented as $(\mathbf{u}^T \parallel \mathbf{v}^T)^T$, which yields a column vector.
- The infinity norm (or ℓ_∞ norm) of a vector is written as $\|\cdot\|_\infty$.
- The function $\text{bin}(\cdot)$ stands for the binary decomposition of an integer or a vector of integers.
- The notation $\text{neg}(\cdot)$ designates a negligible function with respect to the security parameter.

2.1.1 Lattice Hard Problems

Definition 1 Short Integer Solution (SIS) [35, 36]:

The $\text{SIS}_{m,n,q,\beta}$ problem is defined as follows: given a uniformly random matrix $\tilde{\mathbf{B}} \xleftarrow{\$} \mathbb{Z}_q^{n \times m}$, the goal is to find a non-zero vector $\mathbf{s} \in \mathbb{Z}_q^m$ such that

$$\tilde{\mathbf{B}} \cdot \mathbf{s} = \mathbf{0} \pmod{q} \quad \text{and} \quad \|\mathbf{s}\|_\infty \leq \beta.$$

Definition 2 Learning With Rounding (LWR) [37]:

Let $\mathbf{s} \in \mathbb{Z}_q^m$ be a secret vector. The LWR distribution $L_{\mathbf{s}}$ is defined over $\mathbb{Z}_q^m \times \mathbb{Z}_p$ as follows: sample $\mathbf{a} \xleftarrow{\$} \mathbb{Z}_q^m$ uniformly at random and compute

$$b = \lfloor \langle \mathbf{a}, \mathbf{s} \rangle \rfloor_p,$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product and rounding is performed from $\mathbb{Z}_q$ to $\mathbb{Z}_p$. The decision-LWR problem is to distinguish between m_1 independent samples drawn from $L_{\mathbf{s}}$ and m_1 uniformly random samples from $\mathbb{Z}_q^m \times \mathbb{Z}_p$. The advantage of any distinguisher is denoted by $\text{Adv}^{\text{LWR}}(\text{Solver})$.

2.2 Zero-Knowledge Arguments of Knowledge (ZKAoK)

A zero-knowledge argument of knowledge (ZKAoK) [38] enables a prover to demonstrate to a verifier that they hold a valid witness for a particular statement, while revealing nothing except the fact that the statement is true.

Let $\mathcal{R} \subseteq \{0, 1\}^* \times \{0, 1\}^*$ denote a binary relation between statements and corresponding witnesses. A zero-knowledge argument system for $\mathcal{R}$ between a prover $\mathcal{P}$ and a verifier $\mathcal{V}$ satisfies the following properties:

- **Completeness:** For $(s, w) \in \mathcal{R}$, an honest verifier will accept the proof produced by an honest prover with probability 1:

$$\Pr[\mathcal{V} \text{ accepts } \pi \leftarrow \mathcal{P}(s, w)] = 1.$$

- **Soundness:** If $(s, w) \notin \mathcal{R}$, then for any probabilistic polynomial-time (PPT) adversarial prover $\mathcal{P}^*$, the verifier is convinced with negligible probability:

$$\text{Adv}_{\mathcal{P}^*, \mathcal{V}}^{\text{sound}} := \Pr[\mathcal{V} \text{ accepts } \pi \leftarrow \mathcal{P}^*(s)] \leq \varepsilon,$$

where ε is the soundness error.

- **Statistical Zero-Knowledge:** For every PPT verifier $\mathcal{V}^*$, there exists a simulator $\mathcal{S}$ capable of producing a transcript that is statistically indistinguishable from that generated by interaction with an honest prover, thus guaranteeing the statistical zero-knowledge property.

In our construction, we adopt the efficient zero-knowledge protocol proposed by Yang et al. [23], which realizes a ZKAoK for the relation $\mathcal{R}_{\text{ZK}}$ defined as:

$$\mathcal{R}_{\text{ZK}} = \{((\mathbf{A}, \mathbf{z}, \mathcal{M}), \mathbf{x}) \mid \mathbf{A}\mathbf{x} = \mathbf{z} \wedge \forall (i, j, k) \in \mathcal{M}, \mathbf{x}_i = \mathbf{x}_j \cdot \mathbf{x}_k\},$$

where $\mathbf{A}$ and $\mathbf{z}$ are public, $\mathbf{x}$ is the secret witness vector, and $\mathcal{M}$ contains index triples enforcing quadratic constraints. The completeness, soundness, and zero-knowledge properties are formally proven in Theorem 3.1 of [23].

2.3 Signature Proofs of Knowledge (SPK)

A *signature proof of knowledge* (SPK) is obtained by converting an interactive zero-knowledge proof into a non-interactive signature through the Fiat–Shamir heuristic [39]. It allows a signer to prove knowledge of a witness related to a public statement, without leaking any secret information.

Given a relation $\mathcal{R} \subseteq \{0, 1\}^* \times \{0, 1\}^*$, the relation

$$\text{PK}\{(s, w) \mid (s, w) \in \mathcal{R}\}$$

denotes a proof that the prover knows a witness w such that $(s, w) \in \mathcal{R}$.

To make the proof non-interactive and bind it to a message $\mathbf{m}$, the Fiat–Shamir transform replaces the verifier’s challenge with a hash of the protocol and message. The resulting non-interactive proof becomes a signature:

$$\text{SPK}\{(s, w) \mid (s, w) \in \mathcal{R}\} \{\mathbf{m}\}.$$

This SPK serves as both a zero-knowledge proof and a digital signature, enabling secure, privacy-preserving authentication.

2.4 Linkable Ring Signature Scheme

Following the work of [20], the linkable ring signature includes five algorithms $\text{LRS}^+ = (\text{Setup}, \text{UKeygen}, \text{Sign}, \text{Vrfy}, \text{Link})$. The explanation of algorithms is as follows:

- $\text{LRS}^+.\text{Setup}(\lambda)$: On input of the security parameter λ , this algorithm generates and outputs the public parameters pp , which are required for all subsequent procedures.
- $\text{LRS}^+.\text{UKeygen}(\text{pp})$: Generates a key pair $(\text{usk}, \mathbf{p})$, where usk is the secret key and $\mathbf{p}$ is the corresponding public key.
- $\text{LRS}^+.\text{Sign}(\text{pp}, \text{usk}, \mathbf{m}, \mathbf{S}_{pk})$: Takes as input the public parameters pp , a secret key usk associated with one of the public keys in the ring $\mathbf{S}_{pk}$, and a message $\mathbf{m}$, and produces a signature σ .
- $\text{LRS}^+.\text{Vrfy}(\text{pp}, \mathbf{m}, \mathbf{S}_{pk}, \sigma)$: Checks if σ constitutes a valid signature on the message $\mathbf{m}$ for the ring $\mathbf{S}_{pk}$, and returns 1 if valid, or 0 otherwise.
- $\text{LRS}^+.\text{Link}(\text{pp}, \sigma_1, \sigma_2)$: Takes two signatures and outputs 1 if they were generated by the same signer (i.e., linkable), and 0 otherwise.

2.4.1 Correctness and security of LRS^+

Correctness:

The correctness of the linkable ring signature (LRS^+) requires the following two conditions:

- **Verification Correctness.** Let $\text{pp} \leftarrow \text{LRS}^+.\text{Setup}(\lambda)$, $(\text{usk}, \mathbf{p}) \leftarrow \text{LRS}^+.\text{UKeygen}(\text{pp})$. For any message $\mathbf{m}$, and polynomial-size ring $\mathbf{S}_{pk}$ containing public keys $\mathbf{p}$, we have:

$$\Pr [\text{LRS}^+.\text{Vrfy}(\text{pp}, \mathbf{m}, \mathbf{S}_{pk}, \sigma) = 1] \geq 1 - \text{neg}(\lambda).$$

- **Linking Correctness.** Let $\text{pp} \leftarrow \text{LRS}^+.\text{Setup}(\lambda)$, $(\text{usk}_1, \mathbf{p}_1) \leftarrow \text{LRS}^+.\text{UKeygen}(\text{pp})$ and $(\text{usk}_2, \mathbf{p}_2) \leftarrow \text{LRS}^+.\text{UKeygen}(\text{pp})$. For any messages $\mathbf{m}_1, \mathbf{m}_2$, any polynomial-size ring $\mathbf{S}_1$ containing $\mathbf{p}_1$, and $\mathbf{S}_2$ containing $\mathbf{p}_2$, we require:

$$\begin{aligned} &\Pr [\text{LRS}^+.\text{Link}(\text{LRS}^+.\text{Sign}(\text{pp}, \text{usk}_1, \mathbf{m}_1, \mathbf{S}_1), \text{LRS}^+.\text{Sign} \\ &\quad (\text{pp}, \text{usk}_2, \mathbf{m}_2, \mathbf{S}_2)) = 1] \geq 1 - \text{neg}(\lambda), \\ &\Pr [\text{LRS}^+.\text{Link}(\text{LRS}^+.\text{Sign}(\text{pp}, \text{usk}_1, \mathbf{m}_1, \mathbf{S}_1), \text{LRS}^+.\text{Sign} \\ &\quad (\text{pp}, \text{usk}_2, \mathbf{m}_2, \mathbf{S}_2)) = 0] \geq 1 - \text{neg}(\lambda). \end{aligned}$$

Security:

The security of LRS^+ demands that it satisfies unforgeability, linkable-anonymity, linkability, and non-slanderability for adversarially chosen keys. These properties are formally defined as follows using these two oracles:

$\mathcal{O}_{\text{sign}}(j, m, \mathbf{R} \subseteq \mathbf{S}_{pk})$: This oracle returns signature $\sigma \leftarrow \text{Sign}(j, m, \mathbf{R})$, where $\mathbf{p}_j \in \mathbf{R}$. The tuple $(m, \mathbf{R}, \sigma)$ is then stored in the set $\mathbf{SO}$.

$\mathcal{O}_{\text{crpt}}(j)$: This returns ω_j , which is the randomness used in the generation of key pair $(\text{usk}_j, \mathbf{p}_j)$ and adds $\mathbf{p}_j$ to the set $\mathbf{CO}$.

Unforgeability. The LRS^+ is *unforgeable* if for any PPT adversary $\mathcal{A}$, the advantage of $\mathcal{A}$ in the following game is negligible in λ :

1. The challenger executes $\text{LRS}^+.\text{Setup}(\lambda)$ to obtain public parameters pp and generates key pairs $(\text{usk}_j, \mathbf{p}_j) \leftarrow \text{LRS}^+.\text{UKeygen}(\text{pp}; \omega_j)$.
It then sends pp and the public key set $\mathbf{S}_{pk} = \{\mathbf{p}_j\}_{j=1}^N$ to $\mathcal{A}$, and initializes two empty sets: $\mathbf{SO}$ and $\mathbf{CO}$.
2. $\mathcal{A}$ has access to the the oracles $\mathcal{O}_{\text{sign}}$ and $\mathcal{O}_{\text{crpt}}$.
3. Finally, $\mathcal{A}$ outputs $(m^*, \mathbf{R}^*, \sigma^*)$ and wins if all of the conditions hold: $\text{LRS}^+.\text{Vrfy}(\text{pp}, m^*, \mathbf{R}^*, \sigma^*) = 1$, $\mathbf{R}^* \subseteq \mathbf{S}_{pk}$, $\mathbf{R}^* \cap \mathbf{SO} = \emptyset$ and $(m^*, \mathbf{R}^*, \sigma^*) \notin \mathbf{SO}$.

Linkable-Anonymity w.r.t. Adversarially-Chosen Keys. The LRS^+ scheme is said to be linkable-anonymous for adversarially-chosen keys, if for any PPT adversary $\mathcal{A}$, the probability that $\mathcal{A}$ succeeds in the following experiment is negligibly close to $\frac{1}{2}$:

1. The challenger generates the public parameters by running $\text{pp} \leftarrow \text{LRS}^+.\text{Setup}(\lambda)$ and and generates key pairs $(\text{usk}_j, \mathbf{p}_j) \leftarrow \text{LRS}^+.\text{UKeygen}(\text{pp}; \omega_j)$. It then sends pp and the public key set $\mathbf{S}_{pk} = \{\mathbf{p}_j\}_{j=1}^N$ to $\mathcal{A}$, and initializes two empty sets: $\mathbf{SO}$ and $\mathbf{CO}$.
2. The adversary $\mathcal{A}$ is granted access to two oracles $\mathcal{O}_{\text{sign}}$ and $\mathcal{O}_{\text{crpt}}$.
3. The adversary $\mathcal{A}$ submits a message m^* , two distinct indices j_0^*, j_1^* , and a ring $\mathbf{R}^* \subseteq \mathbf{S}_{pk}$ such that $\mathbf{p}_{j_0^*}, \mathbf{p}_{j_1^*} \in \mathbf{R}^*$. The challenger selects a random bit $b \in \{0, 1\}$ and returns the challenge signature $\sigma^* \leftarrow \text{Sign}(m^*, \mathbf{R}^*, \text{usk}_{j_b^*})$.
4. After receiving the challenge signature σ^* , the adversary $\mathcal{A}$ continues to access the signing oracle $\mathcal{O}_{\text{sign}}$ and the corrupt oracle $\mathcal{O}_{\text{crpt}}$ as in the first phase. If $\mathcal{A}$ queries either $(j_0^*, \cdot, \cdot)$ or $(j_1^*, \cdot, \cdot)$ to the signing oracle $\mathcal{O}_{\text{sign}}$, or if j_0^* or j_1^* belongs to the corruption set $\mathbf{CO}$, then $\mathcal{A}$ immediately fails and the game terminates.
5. Finally, $\mathcal{A}$ gives output a bit b' . The adversary wins if and only if $b' = b$.

Linkability w.r.t. adversarially-chosen keys. The LRS^+ scheme is said to be *linkable with respect to adversarially-chosen keys* if for any PPT adversary $\mathcal{A}$, the probability that $\mathcal{A}$ wins in the following game is negligible:

1. The challenger generates the public parameters by running $\text{pp} \leftarrow \text{LRS}^+.\text{Setup}(\lambda)$ and and generates key pairs $(\text{usk}_j, \mathbf{p}_j) \leftarrow \text{LRS}^+.\text{UKeygen}(\text{pp}; \omega_j)$. It then sends pp and the public key set $\mathbf{S}_{pk} = \{\mathbf{p}_j\}_{j=1}^N$ to $\mathcal{A}$, and initializes two empty sets: $\mathbf{SO}$ and $\mathbf{CO}$.
2. The adversary $\mathcal{A}$ can access two oracles $\mathcal{O}_{\text{sign}}$ and $\mathcal{O}_{\text{crpt}}$.
3. Eventually, $\mathcal{A}$ outputs two message-ring-signature tuples: $(m_1^*, \mathbf{R}_1^*, \sigma_1^*)$ and $(m_2^*, \mathbf{R}_2^*, \sigma_2^*)$. The adversary wins if all the conditions are satisfied: $\text{LRS}^+.\text{Vrfy}(\text{pp}, m_1^*, \mathbf{R}_1^*, \sigma_1^*) = 1$, $\text{LRS}^+.\text{Vrfy}(\text{pp}, m_2^*, \mathbf{R}_2^*, \sigma_2^*) = 1$, $\text{LRS}^+.\text{Link}(\text{pp}, \sigma_1^*, \sigma_2^*) = 0$, $(m_1^*, \mathbf{R}_1^*, \sigma_1^*) \notin \mathbf{SO}$, $(m_2^*, \mathbf{R}_2^*, \sigma_2^*) \notin \mathbf{SO}$, and $|(\mathbf{R}_1^* \cup \mathbf{R}_2^*) \cap \mathbf{CO}| + |(\mathbf{R}_1^* \cup \mathbf{R}_2^*) \setminus \mathbf{S}_{pk}| \leq 1$.

Non-slanderability w.r.t. adversarially-chosen keys. The LRS^+ scheme is said to be *non-slanderable with respect to adversarially-chosen keys* if for any PPT adversary $\mathcal{A}$, the advantage of $\mathcal{A}$ in winning the following game is negligible:

1. The challenger generates the public parameters by running $\text{pp} \leftarrow \text{LRS}^+.\text{Setup}(\lambda)$ and and generates key pairs $(\text{usk}_j, \mathbf{p}_j) \leftarrow \text{LRS}^+.\text{UKeygen}(\text{pp}; \omega_j)$. It then sends pp and the public key set $\mathbf{S}_{pk} = \{\mathbf{p}_j\}_{j=1}^N$ to $\mathcal{A}$, and initializes two empty sets: $\mathbf{SO}$ and $\mathbf{CO}$.
2. The adversary $\mathcal{A}$ can access the oracles $\mathcal{O}_{\text{sign}}$ and $\mathcal{O}_{\text{crpt}}$.
3. Finally, the adversary $\mathcal{A}$ gives output $(j, \sigma, m^*, \mathbf{R}^*, \sigma^*)$ and succeeds if all the conditions hold: $\text{LRS}^+.\text{Vrfy}(\text{pp}, m^*, \mathbf{R}^*, \sigma^*) = 1$, $\text{LRS}^+.\text{Link}(\text{pp}, \sigma, \sigma^*) = 1$, $(j, *, *, \sigma) \in \mathbf{SO}$, $j \notin \mathbf{CO}$ and $(j, m^*, \mathbf{R}^*, \sigma^*) \notin \mathbf{SO}$.

2.5 Weak Pseudorandom Function (wPRF)

We describe a weak pseudorandom function (wPRF) based on the Learning With Rounding (LWR) problem, following the methodology presented in [24]. The security of the construction stems from the computational hardness of solving the LWR problem.

Let $\mathcal{F}$ be the function defined over the domain $\mathbb{Z}_q^{m_2 \times m_1}$, with outputs in $\mathbb{Z}_p^{m_2}$, and keyed by elements from $\mathbb{Z}_q^{m_1}$. Here, m_1, m_2, p, q are integers such that $p \geq 2$, and $\gamma' = q/p$ is assumed to be an odd integer. To ensure the function's pseudorandomness, the following relation must be satisfied:

$$m_2 \geq m_1 \cdot \frac{\log q + 1}{\log p - 1}.$$

The function $\mathcal{F}$ comprises the following two core algorithms:

- **Key Generation:** Given a security parameter λ , sample the key $\text{usk} \xleftarrow{\$} \mathbb{Z}_q^{m_1}$ uniformly at random.
- **Evaluation:** For an input matrix $\tilde{\mathbf{T}} \in \mathbb{Z}_q^{m_2 \times m_1}$ and a secret key $\text{usk} \in \mathbb{Z}_q^{m_1}$, the function outputs

$$\mathcal{F}_{\text{usk}}(\tilde{\mathbf{T}}) = \left\lfloor \tilde{\mathbf{T}} \cdot \text{usk} \right\rfloor_p = \left\lfloor \frac{p}{q} \cdot \tilde{\mathbf{T}} \cdot \text{usk} \right\rfloor \bmod p.$$

The security of $\mathcal{F}$ is formalized by the following theorem:

Theorem 1 ([24]) *Considering the hardness of the LWR problem, the function $\mathcal{F}$ is a secure weak pseudorandom function provided that the parameter condition $m_2 \geq m_1 \cdot \frac{\log q + 1}{\log p - 1}$ holds.*

3 The Accumulator Scheme

This section builds upon the frameworks established in [40, 22], introducing a modified Merkle tree accumulator that incorporates the accumulation of a unique tag, denoted as $\tilde{\mathbf{t}} = \text{bin}(\mathcal{F}_{\text{usk}}(\tilde{\mathbf{T}}) \bmod p) \in \{0, 1\}^{m_2 k'_1}$ alongside the public keys of the users in the ring. Since the dimension of $\tilde{\mathbf{t}}$ differs from that of the public keys, the matrices $\mathbf{C}_1, \mathbf{C}_2 \in \mathbb{Z}_q^{n \times m_2 k'_1}$ are introduced to map the tag into the value space of the accumulator. Next, we define the family $\mathcal{H}$ of collision-resistant hash functions, constructed from lattice-based assumptions, which underpin the Merkle tree accumulator. In this setting, the parameters q and n are positive integers with $k_1 = \lceil \log q \rceil$ and $m_1 = 2nk_1$.

Definition 3 *Consider $\mathcal{H}$ be the family of hash functions such that $\mathcal{H} = \{\mathbf{h}_{\tilde{\mathbf{B}}, \tilde{\mathbf{C}}, \tilde{\mathbf{t}}} : \tilde{\mathbf{B}} = [\mathbf{B}_1, \mathbf{B}_2] \in \mathbb{Z}_q^{n \times nk_1} \times \mathbb{Z}_q^{n \times nk_1}, \tilde{\mathbf{C}} = [\mathbf{C}_1, \mathbf{C}_2] \in \mathbb{Z}_q^{n \times m_2 k'_1} \times \mathbb{Z}_q^{n \times m_2 k'_1}, \tilde{\mathbf{t}} \in \{0, 1\}^{m_1 k'_1}\}$, where $\mathbf{h}_{\tilde{\mathbf{B}}, \tilde{\mathbf{C}}, \tilde{\mathbf{t}}}$ is a mapping from $\{0, 1\}^{nk_1} \times \{0, 1\}^{nk_1}$ to $\{0, 1\}^{nk_1}$ and it is defined as follows:*

$$\mathbf{h}_{\tilde{\mathbf{B}}, \tilde{\mathbf{C}}, \tilde{\mathbf{t}}}(\mathbf{u}_0, \mathbf{u}_1) = \text{bin}(\mathbf{B}_1 \cdot \mathbf{u}_0 + \mathbf{B}_2 \cdot \mathbf{u}_1 + \mathbf{C}_1 \cdot \tilde{\mathbf{t}} + \mathbf{C}_2 \cdot \tilde{\mathbf{t}} \bmod q) \in \{0, 1\}^{nk_1}.$$

The above equation can be rewritten as

$$\mathbf{B}_1 \cdot \mathbf{u}_0 + \mathbf{B}_2 \cdot \mathbf{u}_1 + \mathbf{C}_1 \cdot \tilde{\mathbf{t}} + \mathbf{C}_2 \cdot \tilde{\mathbf{t}} = \mathbf{G} \cdot \mathbf{h}_{\tilde{\mathbf{B}}, \tilde{\mathbf{C}}, \tilde{\mathbf{t}}}(\mathbf{u}_0, \mathbf{u}_1) \bmod q$$

where, $\mathbf{G} = \mathbf{I}_n \otimes (1 \ 2 \ 2^2 \ \dots \ 2^{k_1-1})$ is gadget matrix.

Lemma 1 [40] *Consider the hardness of the SIS problem, the family of function $\mathcal{H}$ (in Definition 3) is considered collision-resistant.*

We refer to [40] for the proof of the above lemma.

3.1 Merkle Tree Accumulator

The scheme consists of four algorithms, collectively denoted as $\text{MT} = \{\text{Setup}, \text{Accum}, \text{Witns}, \text{Verify}\}$. This accumulator scheme is designed to accumulate a user's public key with a corresponding unique tag. The functionality of each algorithm is described below:

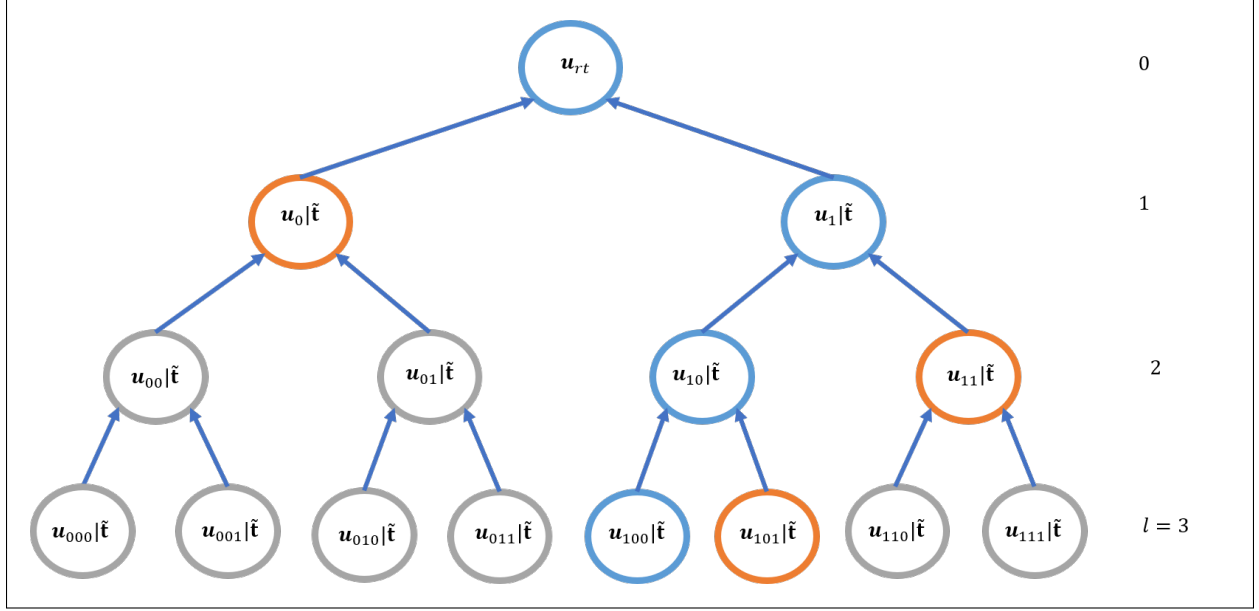


Figure 2: A Merkle tree with eight leaf nodes is constructed, where the leaf node values are $\mathbf{u}_{000}|\tilde{\mathbf{t}}, \dots, \mathbf{u}_{111}|\tilde{\mathbf{t}}$. The node with value $\mathbf{u}_{100}|\tilde{\mathbf{t}}$ is accumulated into $\mathbf{u}_{rt}$. The nodes $\{\mathbf{u}_{100}|\tilde{\mathbf{t}}, \mathbf{u}_{10}|\tilde{\mathbf{t}}, \mathbf{u}_1|\tilde{\mathbf{t}}\}$ and $\{\mathbf{u}_{101}|\tilde{\mathbf{t}}, \mathbf{u}_{11}|\tilde{\mathbf{t}}, \mathbf{u}_0|\tilde{\mathbf{t}}\}$ act as witnesses for the accumulation of $\mathbf{u}_{100}|\tilde{\mathbf{t}}$ into $\mathbf{u}_{rt}$.

Procedure: *Setup and Accumulation in Merkle Tree-based Accumulator Scheme*

- **MT.Setup(λ):** Accepts the security parameter λ and outputs public parameters $\text{pp} = \{\tilde{\mathbf{B}}, \tilde{\mathbf{C}}\}$, where $\tilde{\mathbf{B}} \xleftarrow{\$} \mathbb{Z}_q^{n \times m_1}$ and $\tilde{\mathbf{C}} \xleftarrow{\$} \mathbb{Z}_q^{n \times 2m_2k'_1}$.
- **MT.Accum($\text{pp}, \tilde{\mathbf{t}}, \mathbf{S}_{pk}$):** Let $\mathbf{S}_{pk} = \{\mathbf{p}_0, \dots, \mathbf{p}_{N-1}\} \in \{\{0, 1\}^{nk_1}\}^N$ where $N = 2^l$. Build a Merkle tree as follows:
 - (a) Assign leaf $\mathbf{u}_{j_1, \dots, j_l} \leftarrow \mathbf{p}_j$, where $(j_1, \dots, j_l)$ is binary representation of j .
 - (b) For $i \in [l-1]$, set:

$$\mathbf{u}_{j'_1, \dots, j'_i} = \mathbf{h}_{\tilde{\mathbf{B}}, \tilde{\mathbf{C}}, \tilde{\mathbf{t}}}(\mathbf{u}_{j'_1, \dots, j'_i, 0}, \mathbf{u}_{j'_1, \dots, j'_i, 1})$$
 - (c) Set accumulated value: $\mathbf{u}_{rt} = \mathbf{h}_{\tilde{\mathbf{B}}, \tilde{\mathbf{C}}, \tilde{\mathbf{t}}}(\mathbf{u}_0, \mathbf{u}_1)$.

Output $\mathbf{u}_{rt}$.

- **MT.Witns($\text{pp}, \tilde{\mathbf{t}}, \mathbf{p}$):** If $\mathbf{p} \notin \mathbf{S}_{pk}$ return $\perp$. Otherwise, for some j , $\mathbf{p}_j = \mathbf{p}$. Output witness: $\text{wit} = \{(j_1, \dots, j_l), \{\mathbf{v}_i\}_{i=1}^l, \{\mathbf{w}_i\}_{i=1}^l\} \in \{0, 1\}^l \times (\{0, 1\}^{nk_1})^l \times (\{0, 1\}^{nk_1})^l$.
- **MT.Verify($\text{pp}, \mathbf{u}_{rt}, \mathbf{p}, \tilde{\mathbf{t}}, \text{wit}$):** The verifier performs:

- Set $\mathbf{v}_l = \mathbf{p}$
- For $i = l-1$ to 0, recursively compute:

$$\mathbf{v}_i = \begin{cases} \mathbf{h}_{\tilde{\mathbf{B}}, \tilde{\mathbf{C}}, \tilde{\mathbf{t}}}(\mathbf{w}_{i+1}, \mathbf{v}_{i+1}) & \text{if } \bar{j}_{i+1} = 0 \\ \mathbf{h}_{\tilde{\mathbf{B}}, \tilde{\mathbf{C}}, \tilde{\mathbf{t}}}(\mathbf{v}_{i+1}, \mathbf{w}_{i+1}) & \text{if } \bar{j}_{i+1} = 1 \end{cases} \quad (1)$$

- Return 1 if $\mathbf{v}_0 = \mathbf{u}_{rt}$, else return 0.

In Figure 2, we present a demonstration of a Merkle tree accumulator scheme with eight leaf nodes.

Definition 4 (Security:)[22] *The Merkle Tree Accumulator scheme, $MT = \{\text{Setup}, \text{Accum}, \text{Witns}, \text{Verify}\}$ is secure if the following condition holds for any PPT adversary $\mathcal{A}$,*

$$\begin{aligned} \Pr[\text{pp} \leftarrow MT.\text{Setup}(\lambda); (\mathbf{S}_{pk}, \mathbf{p}^*, \mathbf{wit}^*) \leftarrow \mathcal{A}(\text{pp}) : \mathbf{p}^* \notin \mathbf{S}_{pk} \\ \wedge MT.\text{Verify}(\text{pp}, MT.\text{Accum}(\text{pp}, \tilde{\mathbf{t}}, \mathbf{S}_{pk}), \mathbf{p}^*, \tilde{\mathbf{t}}, \mathbf{wit}^*) = 1] \\ = \text{neg}(\lambda). \end{aligned}$$

Theorem 2 *The Merkle Tree Accumulator (MT) is secure, as defined in Definition 4, under the hardness assumption of the SIS problem.*

Proof 1 We begin with the assumption that an adversary $\mathcal{A}$ has a non-negligible success probability in the context of Definition 4. Based on this, we construct an algorithm $\mathcal{B}$ that solves an instance of the SIS problem with non-negligible probability. Now, $\mathcal{B}$ receives output $\tilde{\mathbf{B}} \in \mathbb{Z}_q^{n \times m_1}$, $\tilde{\mathbf{C}} \in \mathbb{Z}_q^{n \times 2m_2k'_1}$ from $MT.\text{Setup}(\lambda)$ and send it to $\mathcal{A}$. $\mathcal{A}$ receives pp and returns $(\mathbf{S}_{pk}, \mathbf{p}^*, \mathbf{wit}^*)$ such that

$$MT.\text{Verify}(\text{pp}, \mathbf{u}_{root}^*, \mathbf{p}^*, \tilde{\mathbf{t}}, \mathbf{wit}^*) = 1$$

with $\mathbf{p}^* \notin \mathbf{S}_{pk}$ and $\mathbf{u}_{rt}^* = MT.\text{Accum}(\text{pp}, \tilde{\mathbf{t}}, \mathbf{S}_{pk})$.

The witness $\mathbf{wit} = \{(j_1^*, j_2^*, \dots, j_l^*), \{\mathbf{v}_i^*\}_{i \in [1, l]}, \{\mathbf{w}_i^*\}_{i \in [1, l]}\}$, where $(j_1^*, j_2^*, \dots, j_l^*)$ is the binary representation of $j^* \in [0, N - 1]$. Now, consider the path from $\mathbf{p}_{j^*}$ to $\mathbf{u}_{rt}^*$ such that $\mathbf{u}_{j_1^*, j_2^*, \dots, j_l^*}^* = \mathbf{p}_{j^*}, \mathbf{u}_{j_1^*, j_2^*, \dots, j_{l-1}^*}^*, \dots, \mathbf{u}_{j_1^*}^*, \mathbf{u}_{rt}^*$. On the other side, the path computed by $MT.\text{Verify}(\text{pp}, \mathbf{u}_{rt}^*, \mathbf{p}^*, \tilde{\mathbf{t}}, \mathbf{wit}^*)$ is $\mathbf{v}_l^* = \mathbf{p}^*, \mathbf{v}_{l-1}^*, \dots, \mathbf{v}_1^*, \mathbf{v}_0^* = \mathbf{u}_{rt}^*$. As $\mathbf{p}^* \notin \mathbf{S}_{pk}$, it follows that $\mathbf{p}^* \neq \mathbf{p}_{j^*}$, which leads to finding the smallest integer $b \in [l]$ such that $\mathbf{v}_b^* \neq \mathbf{u}_{j_1^*, j_2^*, \dots, j_b^*}^*$. Therefore, a collision is obtained in the hash function $\mathbf{h}_{\tilde{\mathbf{B}}, \tilde{\mathbf{T}}, \tilde{\mathbf{t}}}$ at the parent node of $\mathbf{u}_{j_1^*, j_2^*, \dots, j_b^*}^*$. The proof of the theorem then follows from Lemma 1.

4 Our Linkable Ring Signature

Our linkable ring signature scheme LRS^+ is defined as a tuple of five algorithms: (Setup, UKeygen, Sign, Vrfy, Link). This scheme employs a Merkle tree accumulator to verify the signer's membership in the ring. The unique tag, which is derived from the binary decomposition of wPRF output i.e. $\text{bin}(\mathcal{F}_{\text{usk}}(\tilde{\mathbf{T}}))$, where usk is the signer's secret key. These algorithms work as follows:

- $\text{LRS}^+.\text{Setup}(\lambda)$: Given the security parameter λ as input, this algorithm outputs the public parameters.

$$\text{pp} = \{n, m_1, q, p, k_1, k'_1, m_2, m', \gamma', \tilde{\mathbf{B}}, \tilde{\mathbf{C}}\}$$

where,

- $n = \text{poly}(\lambda)$ is a positive integer.
- $p \geq 2$ and q are positive integer, $\gamma' = q/p$ is positive odd integer, $k'_1 = \lceil \log p \rceil$.
- $m_1 = 2nk_1$, where $k_1 = \lceil \log q \rceil$.
- $m_2 \geq m_1 \cdot (\log q + 1)/(\log p - 1)$ is positive integer and $m' = m_2k'_1$.
- $\tilde{\mathbf{B}} = [\mathbf{B}_1 | \mathbf{B}_2] \xleftarrow{\$} \mathbb{Z}_q^{n \times nk_1} \times \mathbb{Z}_q^{n \times nk_1}$.
- $\tilde{\mathbf{T}} \xleftarrow{\$} \mathbb{Z}_q^{m_2 \times m_1}$.
- $\tilde{\mathbf{C}} = [\mathbf{C}_1 | \mathbf{C}_2] \xleftarrow{\$} \mathbb{Z}_q^{n \times m_2k'_1} \times \mathbb{Z}_q^{n \times m_2k'_1}$ is the matrix for the accumulator scheme.
- $\text{LRS}^+.\text{UKeygen}(\text{pp})$: This algorithm generates the user's key by first sampling the private key $\text{usk} \xleftarrow{\$} \{0, 1\}^{m_1}$ and computes the public key $\mathbf{p} = \text{bin}(\tilde{\mathbf{B}} \cdot \text{usk} \bmod q) \in \{0, 1\}^{nk_1}$. This outputs the key pair $(\text{usk}, \mathbf{p})$.
- $\text{LRS}^+.\text{Sign}(\text{pp}, \text{usk}, \mathbf{m}, \mathbf{S}_{pk})$: The signer whose public key $\mathbf{p} \in \mathbf{S}_{pk}$, where $\mathbf{S}_{pk} = \{\mathbf{p}_0, \mathbf{p}_1, \dots, \mathbf{p}_{N-1}\}$, executes the following steps to generate a signature on the message $\mathbf{m}$:
 - Generate the unique tag, $\tilde{\mathbf{t}} = \text{bin}(\mathcal{F}_{\text{usk}}(\tilde{\mathbf{T}}) \bmod p) \in \{0, 1\}^{m_2k'_1}$ using wPRF.
 - Evaluate the accumulator value

$$\mathbf{u}_{rt} \leftarrow MT.\text{Accum}(\text{pp}, \tilde{\mathbf{t}}, \mathbf{S}_{pk}).$$

- Calculate the witness:

$$\mathbf{wit} = ((j_1, j_2, \dots, j_l), \{\mathbf{v}_i\}_{i \in [1, l]}, \{\mathbf{w}_i\}_{i \in [1, l]}) \leftarrow \text{MT.Witns}(\text{pp}, \mathbf{p}, \tilde{\mathbf{t}}, \mathbf{S}_{pk}).$$

- Construct a NIZKAoK proof π such that

$$\begin{aligned} \pi = \text{SPK} \Big\{ & (\tilde{\mathbf{B}}, \tilde{\mathbf{C}}, \tilde{\mathbf{T}}, \mathbf{u}_{rt}, \tilde{\mathbf{t}}) \in \mathbb{Z}_q^{n \times m_1} \times \mathbb{Z}_q^{n \times 2m'} \times \mathbb{Z}_q^{m_2 \times m_1} \\ & \times \{0, 1\}^{nk_1} \times \{0, 1\}^{m_2 k'_1}; (\text{usk}, \mathbf{p}, ((j_1, j_2, \dots, j_l), \\ & \{\mathbf{v}_i\}_{i \in [1, l]}, \{\mathbf{w}_i\}_{i \in [1, l]})) \in \{0, 1\}^{m_1} \times \{0, 1\}^{nk_1} \times \\ & \{0, 1\}^l \times (\{0, 1\}^{nk_1})^l \times (\{0, 1\}^{nk_1})^l : \\ & \tilde{\mathbf{B}} \cdot \text{usk} = \mathbf{G} \cdot \mathbf{p} \wedge \mathcal{F}_{\text{usk}}(\tilde{\mathbf{T}}) = \mathbf{G}' \cdot \tilde{\mathbf{t}} \wedge \\ & \text{MT.Verify}(\text{pp}, \mathbf{u}_{rt}, \mathbf{p}, \tilde{\mathbf{t}}, \mathbf{wit}) = 1 \Big\} [m] \end{aligned}$$

where SPK refers to Signature Proof of Knowledge, $\mathbf{G} = \mathbf{I}_n \otimes (1 \ 2 \ 2^2 \ \dots \ 2^{k_1-1})$ and

$$\mathbf{G}' = \mathbf{I}_{m_2} \otimes (1 \ 2 \ 2^2 \ \dots \ 2^{k'_1-1}).$$

A detailed explanation of the proof is provided in the Section 5.

- This algorithm outputs the signature $\sigma = (\pi, \tilde{\mathbf{t}})$.

- $\text{LRS}^+.\text{Vrfy}(\text{pp}, \mathbf{m}, \mathbf{S}_{pk}, \sigma)$: The verifier runs the algorithm to verify the signature $\sigma = (\pi, \tilde{\mathbf{t}})$ by following these steps:
 1. Generate the accumulator value $\mathbf{u}_{rt} \leftarrow \text{MT.Accum}(\text{pp}, \tilde{\mathbf{t}}, \mathbf{S}_{pk})$.
 2. Verify the proof π using accumulator value $\mathbf{u}_{rt}$ and unique tag $\tilde{\mathbf{t}}$.
 3. Return 1 if the verification succeeds; otherwise, return 0.
- $\text{LRS}^+.\text{Link}(\text{pp}, \sigma_1, \sigma_2)$: This algorithm takes as input two signatures $\sigma_1 = (\pi_1, \tilde{\mathbf{t}}_1)$ and $\sigma_2 = (\pi_2, \tilde{\mathbf{t}}_2)$. It outputs 1 iff $\tilde{\mathbf{t}}_1 = \tilde{\mathbf{t}}_2$, indicating that both signatures were likely generated by the same signer.

5 Zero Knowledge Argument of Knowledge For Our Scheme

This section introduces the zero-knowledge arguments, following the framework of [23], for the various relations employed in our signature scheme. A detailed explanation is provided below.

5.1 ZKAoK for $\tilde{\mathbf{B}} \cdot \text{usk} = \mathbf{G} \cdot \mathbf{p} \bmod q$

The linear relation between the signer's key pair is defined as

$$\tilde{\mathbf{B}} \cdot \text{usk} = \mathbf{G} \cdot \mathbf{p} \bmod q,$$

where $\tilde{\mathbf{B}} \in \mathbb{Z}^{n \times m_1}$, $\mathbf{G} \in \mathbb{Z}_q^{n \times nk_1}$ denotes the gadget matrix, $\mathbf{p} \in \{0, 1\}^{nk_1}$ represents the public key, and $\text{usk} \in \{0, 1\}^{m_1}$ denotes the secret key. To ensure the signer's anonymity, the prover constructs a ZKP of the relation, embedding the public key within the secret vector. This relation is reformulated as $\mathbf{A} \cdot \mathbf{x} = \mathbf{z} \bmod q$ with

$$\mathbf{A} = (\tilde{\mathbf{B}} \ -\mathbf{G}), \mathbf{x} = \begin{pmatrix} \text{usk} \\ \mathbf{p} \end{pmatrix} \text{ and } \mathbf{z} = \mathbf{0}^n.$$

The secret vector $\mathbf{x}$ has a length $nk_1 + m_1$ and the size of the constraints set $\mathcal{M}$ is $nk_1 + m_1$.

5.2 ZKAoK for $\mathcal{F}_{\text{usk}}(\tilde{\mathbf{T}}) = \mathbf{G}' \cdot \tilde{\mathbf{t}} \bmod p$

The relation $\mathcal{F}_{\text{usk}}(\tilde{\mathbf{T}}) = \mathbf{G}' \cdot \tilde{\mathbf{t}} \bmod p$ arises from the construction of a unique tag using the wPRF as described in [24].

In this setting, the matrix $\tilde{\mathbf{T}} \in \mathbb{Z}_q^{m_2 \times m_1}$, the gadget matrix $\mathbf{G}' \in \mathbb{Z}^{m_2 \times m_2 k'_1}$ and the unique tag $\tilde{\mathbf{t}} \in \{0, 1\}^{m_2 k'_1}$ are public while the secret key $\text{usk} \in \{0, 1\}^{m_1}$ is private. This relation can be equivalently expressed in the linear form as:

$$[\tilde{\mathbf{T}} \cdot \text{usk}]_p = \mathbf{G}' \cdot \tilde{\mathbf{t}} \bmod p.$$

This is equivalent to

$$\tilde{\mathbf{T}} \cdot \text{usk} + \mathbf{e} = \gamma'(\mathbf{G}' \cdot \tilde{\mathbf{t}}) \bmod q, \quad (2)$$

where $\gamma' = q/p$ and $\mathbf{e} \in [-\beta_1, \beta_1]^{m_2}$ with $\beta_1 = \lfloor \gamma'/2 \rfloor$.

Using the decomposition technique from [41], the secret vector $\mathbf{e}$ decomposed into the binary vector $\bar{\mathbf{e}}$. Let $k = \lfloor \beta_1 \rfloor + 1$ and compute the row vector

$$\mathbf{g}_1 = (\lfloor (\beta_1 + 1)/2 \rfloor \quad \lfloor (\beta_1 + 2)/2^2 \rfloor \quad \dots \quad \lfloor (\beta_1 + 2^{k-1})/2^k \rfloor)$$

such that $\mathbf{H} \cdot \bar{\mathbf{e}} = \mathbf{e}$, where $\mathbf{H} = \mathbf{I}_{m_2} \otimes \mathbf{g}_1$ is a gadget matrix, $\bar{\mathbf{e}} \in \{0, 1\}^{m_2 k}$ and $\mathbf{e} \in \mathbb{Z}_q^{m_2}$. Now, the equation (2) transforms into the format $\mathbf{A} \cdot \mathbf{x} = \mathbf{z} \bmod q$ within the framework of [23] where

$$\mathbf{A} = (\tilde{\mathbf{T}} \quad \mathbf{H}), \mathbf{x} = \begin{pmatrix} \text{usk} \\ \bar{\mathbf{e}} \end{pmatrix} \text{ and } \mathbf{z} = (\gamma'(\mathbf{G}' \cdot \tilde{\mathbf{t}})).$$

The secret vector $\mathbf{x}$ has length $m_1 + m_2 k$, and since vector $\mathbf{x}$ is binary, the $\mathcal{M}$'s size is $m_1 + m_2 k$.

5.3 ZKAoK for Merkle Tree Accumulator

The zero-knowledge argument for the accumulated value in the Merkle Tree accumulator is derived based on the approach outlined in Section 3.1. Equation (1) can be reformulated as follows:

$$\begin{aligned} \mathbf{v}_{i-1} &= \bar{j}_i \cdot \mathbf{h}_{\bar{\mathbf{B}}, \bar{\mathbf{C}}, \tilde{\mathbf{t}}}(\mathbf{v}_i, \mathbf{w}_i) + j_i \cdot \mathbf{h}_{\bar{\mathbf{B}}, \bar{\mathbf{C}}, \tilde{\mathbf{t}}}(\mathbf{w}_i, \mathbf{v}_i) \\ \implies \mathbf{G} \cdot \mathbf{v}_{i-1} &= (1 - j_i)(\mathbf{B}_1 \cdot \mathbf{v}_i + \mathbf{B}_2 \cdot \mathbf{w}_i) + j_i(\mathbf{B}_1 \cdot \mathbf{w}_i + \mathbf{B}_2 \cdot \mathbf{v}_i) \\ &\quad + \mathbf{C}_1 \cdot \tilde{\mathbf{t}} + \mathbf{C}_2 \cdot \tilde{\mathbf{t}} \end{aligned}$$

$\forall i \in \{l, \dots, 1\}$. In order to establish the signer's membership in the set while preserving anonymity, these equations are proven using zero-knowledge techniques. The relation $\mathbf{R}_{\text{MT}}$ is

$$\begin{aligned} \mathbf{R}_{\text{MT}} = \left\{ (\mathbf{B}_1, \mathbf{B}_2, \mathbf{C}_1, \mathbf{C}_2, \mathbf{u}_{rt}, \tilde{\mathbf{t}}), ((j_1, j_2, \dots, j_l), \{\mathbf{v}_i\}_{i \in [1, l]}, \right. \\ \left. \{\mathbf{w}_i\}_{i \in [1, l]}) \in (\mathbb{Z}_q^{n \times nk_1} \times \mathbb{Z}_q^{n \times nk_1} \times \mathbb{Z}_q^{n \times m_2 k'_1} \times \mathbb{Z}_q^{n \times m_2 k'_1} \right. \\ \left. \times \{0, 1\}^{nk_1} \times \{0, 1\}^{m_2 k'_1}) \times (\{0, 1\}^l \times (\{0, 1\}^{nk_1})^l \right. \\ \left. \times (\{0, 1\}^{nk_1})^l) : \right. \\ \mathbf{B}_{1+l_1} \cdot \mathbf{v}_1 + \mathbf{B}_{2-l_1} \cdot \mathbf{w}_1 + \mathbf{C}_1 \cdot \tilde{\mathbf{t}} + \mathbf{C}_2 \cdot \tilde{\mathbf{t}} = \mathbf{G} \cdot \mathbf{u}_{rt} \wedge \\ \forall i \in [2, l], \mathbf{B}_{1+l_i} \cdot \mathbf{v}_i + \mathbf{B}_{2-l_i} \cdot \mathbf{w}_i + \mathbf{C}_1 \cdot \tilde{\mathbf{t}} + \mathbf{C}_2 \cdot \tilde{\mathbf{t}} \\ \left. = \mathbf{G} \cdot \mathbf{v}_{i-1} \right\} \end{aligned}$$

where, $\mathbf{G} = \mathbf{I}_n \otimes (1 \quad 2 \quad 2^2 \quad \dots \quad 2^{k_1-1})$.

To verify the accumulated value using the equations in the relation $\mathbf{R}_{\text{MTA}}$, one must apply the appropriate equation at each level, depending on the position of the witness node

$$\begin{aligned} \mathbf{B}_1 \cdot \mathbf{v}_i + \mathbf{B}_2 \cdot \mathbf{w}_i + \mathbf{C}_1 \cdot \tilde{\mathbf{t}} + \mathbf{C}_2 \cdot \tilde{\mathbf{t}} &= \mathbf{G} \cdot \mathbf{v}_{i-1}, \\ \mathbf{B}_1 \cdot \mathbf{w}_i + \mathbf{B}_2 \cdot \mathbf{v}_i + \mathbf{C}_1 \cdot \tilde{\mathbf{t}} + \mathbf{C}_2 \cdot \tilde{\mathbf{t}} &= \mathbf{G} \cdot \mathbf{v}_{i-1}. \end{aligned}$$

Each component can be expressed as a subset-sum of two separate equations, where the coefficients are assigned as $(1 - j_i, j_i)$.

1. We define, for all $i \in [2, l]$,

$$\begin{aligned} \mathbf{z}_{i,0} &= \mathbf{B}_1 \cdot \mathbf{v}_i + \mathbf{B}_2 \cdot \mathbf{w}_i + \mathbf{C}_1 \cdot \tilde{\mathbf{t}} + \mathbf{C}_2 \cdot \tilde{\mathbf{t}} - \mathbf{G} \cdot \mathbf{v}_{i-1}, \\ \mathbf{z}_{i,1} &= \mathbf{B}_1 \cdot \mathbf{w}_i + \mathbf{B}_2 \cdot \mathbf{v}_i + \mathbf{C}_1 \cdot \tilde{\mathbf{t}} + \mathbf{C}_2 \cdot \tilde{\mathbf{t}} - \mathbf{G} \cdot \mathbf{v}_{i-1}, \\ \mathbf{z}'_{i,0} &= (1 - j_i) \cdot \mathbf{z}_{i,0}, \quad \mathbf{z}'_{i,1} = j_i \cdot \mathbf{z}_{i,1}. \end{aligned}$$

Also, we set

$$\begin{aligned} \mathbf{z}_{1,0} &= \mathbf{B}_1 \cdot \mathbf{v}_1 + \mathbf{B}_2 \cdot \mathbf{w}_1 + \mathbf{C}_1 \cdot \tilde{\mathbf{t}} + \mathbf{C}_2 \cdot \tilde{\mathbf{t}}, \\ \mathbf{z}_{1,1} &= \mathbf{B}_1 \cdot \mathbf{w}_1 + \mathbf{B}_2 \cdot \mathbf{v}_1 + \mathbf{C}_1 \cdot \tilde{\mathbf{t}} + \mathbf{C}_2 \cdot \tilde{\mathbf{t}} \\ \mathbf{z}'_{1,0} &= (1 - j_1) \cdot \mathbf{z}_{1,0}, \quad \mathbf{z}'_{1,1} = j_1 \cdot \mathbf{z}_{1,1}. \end{aligned}$$

2. We take

$$\begin{aligned}\tau_0 &= (1 - j_1 \ 1 - j_2 \dots 1 - j_l)^T, \tau_1 = (j_1 \ j_2 \dots j_l)^T \\ \hat{\mathbf{z}}'_0 &= (\mathbf{z}'_{1,0} \parallel \mathbf{z}'_{2,0} \parallel \dots \parallel \mathbf{z}'_{l,0})^T, \hat{\mathbf{z}}'_1 = (\mathbf{z}'_{1,1} \parallel \mathbf{z}'_{2,1} \parallel \dots \parallel \mathbf{z}'_{l,1})^T \\ \hat{\mathbf{z}}_0 &= (\mathbf{z}_{1,0}^T \parallel \mathbf{z}_{2,0}^T \parallel \dots \parallel \mathbf{z}_{l,0}^T)^T, \hat{\mathbf{z}}_1 = (\mathbf{z}_{1,1}^T \parallel \mathbf{z}_{2,1}^T \parallel \dots \parallel \mathbf{z}_{l,1}^T)^T \\ \hat{\mathbf{v}} &= (\mathbf{v}_1^T \parallel \mathbf{v}_2^T \parallel \dots \parallel \mathbf{v}_l^T)^T, \hat{\mathbf{w}} = (\mathbf{w}_1^T \parallel \mathbf{w}_2^T \parallel \dots \parallel \mathbf{w}_l^T)^T \\ \hat{\mathbf{id}} &= \{\hat{\mathbf{t}}^T \parallel \hat{\mathbf{t}}^T \parallel \dots \parallel \hat{\mathbf{t}}^T\}.\end{aligned}$$

Moreover, we set $\hat{\mathbf{u}} = ((\mathbf{G} \cdot \mathbf{u}_{rt})^T \parallel \mathbf{0}^{1 \times (l-1) \cdot n})^T$.

3. Using the information described above, we construct the public parameters $\mathbf{A}$ and $\mathbf{z}$, along with the secret witness $\mathbf{x}$, corresponding to the accumulated value. These are employed to establish the ZKAoK for the relation $\mathbf{A} \cdot \mathbf{x} = \mathbf{z}$,

$$\begin{aligned}\mathbf{A} &= \begin{pmatrix} \mathbf{I}_l & \mathbf{I}_l & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_{nl} & \mathbf{I}_{nl} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_{nl} & \mathbf{0} & \mathbf{D}_1 & \mathbf{J}_2 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_{nl} & \mathbf{D}_2 & \mathbf{J}_1 \end{pmatrix} \\ \mathbf{x} &= (\tau_0^T \ \tau_1^T \ \hat{\mathbf{z}}_0^T \ \hat{\mathbf{z}}_1^T \ \hat{\mathbf{z}}_0^T \ \hat{\mathbf{z}}_1^T \ \hat{\mathbf{v}}^T \ \hat{\mathbf{w}}^T)^T, \\ \mathbf{z} &= (\mathbf{1}^l \ \hat{\mathbf{u}}^T \ \mathbf{H}_1 \cdot \hat{\mathbf{id}} \ \mathbf{H}_1 \cdot \hat{\mathbf{id}})^T\end{aligned}$$

where,

$$\begin{aligned}\mathbf{D}_1 &= \begin{pmatrix} -\mathbf{B}_1 & & & \\ \mathbf{G} & -\mathbf{B}_1 & & \\ & \ddots & \ddots & \\ & & \mathbf{G} & -\mathbf{B}_1 \end{pmatrix}, \quad \mathbf{D}_2 = \begin{pmatrix} -\mathbf{B}_2 & & & \\ \mathbf{G} & -\mathbf{B}_2 & & \\ & \ddots & \ddots & \\ & & \mathbf{G} & -\mathbf{B}_2 \end{pmatrix}, \\ \mathbf{J}_1 &= -\mathbf{I}_l \otimes \mathbf{B}_1, \quad \mathbf{J}_2 = -\mathbf{I}_l \otimes \mathbf{B}_2, \quad \mathbf{H}_1 = \mathbf{I}_l \otimes (\mathbf{C}_1 + \mathbf{C}_2).\end{aligned}$$

4. We define the sets $\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3$, and $\mathcal{M}_4$ as follows:

$$\begin{aligned}\mathcal{M}_1 &= \{(i, i, i) \mid i \in [1, l]\}, \\ \mathcal{M}_2 &= \{(i, i, i) \mid i \in [2l + 4nl + 1, 2l + 4nl + 2nk_1l]\}, \\ \mathcal{M}_3 &= \{(2l + n(i-1) + j', 2l + 2nl + n(i-1) + j', i) \\ &\quad \mid i \in [1, l], j' \in [1, n]\}, \\ \mathcal{M}_4 &= \{(2l + nl + n(i-1) + j', 2l + 3nl + n(i-1) + j', l + i) \\ &\quad \mid i \in [1, l], j' \in [1, n]\}.\end{aligned}$$

Here: $\mathcal{M}_1$ indicates that τ_0 is a binary vector, $\mathcal{M}_2$ specifies that $\hat{\mathbf{v}}$ and $\hat{\mathbf{w}}$ are binary vectors, for $i \in [1, l]$, $\mathcal{M}_3$ corresponds to $\mathbf{z}'_{i,0} = (1 - j_i) \cdot \mathbf{z}_{i,0}$ and $\mathcal{M}_4$ corresponds to $\mathbf{z}'_{i,1} = j_i \cdot \mathbf{z}_{i,1}$.

Therefore:

$$\mathcal{M} = \mathcal{M}_1 \cup \mathcal{M}_2 \cup \mathcal{M}_3 \cup \mathcal{M}_4.$$

In this relation, the witness length is

$$l(2 + 4n + 2nk_1),$$

and the size of $\mathcal{M}$ is

$$l(1 + 2n + 2nk_1).$$

6 Efficiency of LRS⁺

This section presents an efficiency analysis of the proposed linkable ring signature scheme, denoted as LRS⁺. The selection of parameters is guided by the security requirements of zero-knowledge proof [23], our constructed protocol and the underlying weak pseudorandom function wPRF[24].

In the proposed scheme, the signature size determines the communication overhead, which is influenced by both the proof size and the unique tag. The proof is composed of the following three components:

1. a proof of knowledge linking the signer's secret key to its corresponding public key (Section 5.1).
2. an argument of knowledge for the weak pseudorandom function relation (Section 5.2).
3. zero-knowledge argument for the accumulator scheme (Section 5.3).

Note that in the resultant matrix, the lengths of the reused vectors $\mathbf{usk}$ and $\mathbf{p}$ are counted only once, and therefore, contribute just once to the total signature size.

The overall length of the secret vector is

$$\eta = l(2 + 4n + 2nk_1) + m_1 + m_2k.$$

Also, the total size of $\mathcal{M}$ is

$$\zeta = l(1 + 2n + 2nk_1) + m_1 + m_2k.$$

Asymptotic Complexity of η and ζ : The parameter $l = \log N$, where N denotes the number of users in the ring. Here, $n = \text{poly}(\lambda)$, $k_1 = \log q$ is small, $m_1 = 2nk_1$, and k also depends on $\mathcal{O}(n)$. Since $\log q$ is small, we have $m_1 = \mathcal{O}(n)$ and m_2 depends on n and $\log q$. The dominating term is therefore

$$l(2 + 4n + 2nk_1) = \mathcal{O}(l \cdot n) = \mathcal{O}(\log N \cdot \text{poly}(\lambda)).$$

Thus, the overall complexity of η can be expressed as

$$\eta = \mathcal{O}(\log N \cdot \text{poly}(\lambda)) + \mathcal{O}(\text{poly}(\lambda)). \quad (3)$$

For a fixed security parameter λ , the complexity simplifies to

$$\eta = \Theta(\log N),$$

showing that η grows logarithmically with the ring size, and similarly for ζ .

Signature Size analysis with Respect to ZKP[23]

The size of the non-interactive zero-knowledge argument of knowledge (NIZKAoK) is derived following the structure described in Section 3.2 of [23]. We are employing the parameters such as $\hat{p}$ denotes a positive integer, while $\hat{l}_1$ and $\hat{l}_2$ are parameters associated with the underlying homomorphic commitment scheme employed in the ZK protocol [23]. The parameter Λ specifies the number of repetitions of NIZK, n represents the lattice dimension, q is the modulus, $\hat{k} = 128$; k'_1 , k , and m_2 are parameters used in the wPRF [24]. We get:

$$\|\pi\| = (\log(2\hat{p} + 1) + (3\hat{l}_1 + 2\hat{l}_2 + 2\zeta + 2\eta) \cdot k + \hat{k}) \cdot \Lambda + (\eta + \hat{l}_1) \cdot k \text{ bits.}$$

The signature size is obtained by adding the communication cost of the proof to that of the unique tag, i.e.,

$$\begin{aligned} \|\sigma\| = & (\log(2\hat{p} + 1) + (3\hat{l}_1 + 2\hat{l}_2 + 2\zeta + 2\eta) \cdot k + \hat{k}) \cdot \Lambda + (\eta + \hat{l}_1) \cdot k \\ & + m_1k' \text{ bits.} \end{aligned}$$

For 80-bit security, a small root-Hermite factor of 1.0048 is needed (as discussed in Section B.1, [23]) to compromise the underlying assumptions. The associated parameters are chosen as follows: $\hat{p} = 2^{40}$, $l = 10$, $\hat{l}_1 = 3700$, $\hat{l}_2 = 4000$, $\Lambda = 2$, $n = 10$, $k_1 = 120$, $k'_1 = 118$, $k = 2$, $m_2 = 2483$ and the signature size is $\|\sigma\| = 4.7$ MB.

For a 100-bit security level, the small root-Hermite factor reduces to 1.0042. The chosen parameters are: $\hat{p} = 2^{50}$, $l = 10$, $\hat{l}_1 = 5100$, $\hat{l}_2 = 5400$, $\Lambda = 2$, $n = 10$, $k_1 = 140$, $k'_1 = 138$, $k = 2$, $m_2 = 2882$. The signature size is $\|\sigma\| = 6.54$ MB.

For a 128-bit security level, a root-Hermite factor of 1.0035 is required to compromise the assumptions. The parameters are set as: $\hat{p} = 2^{64}$, $l = 10$, $\hat{l}_1 = 7400$, $\hat{l}_2 = 7900$, $\Lambda = 2$, $n = 11$, $k_1 = 170$, $k'_1 = 168$, $k = 2$, $m_2 = 3830$ and leading to a signature size of $\|\sigma\| = 10.73$ MB.

Figure 3 illustrates the signature sizes computed for security levels of 80, 100, and 128-bits, evaluated for different ring sizes (i.e., varying numbers of users in the ring).

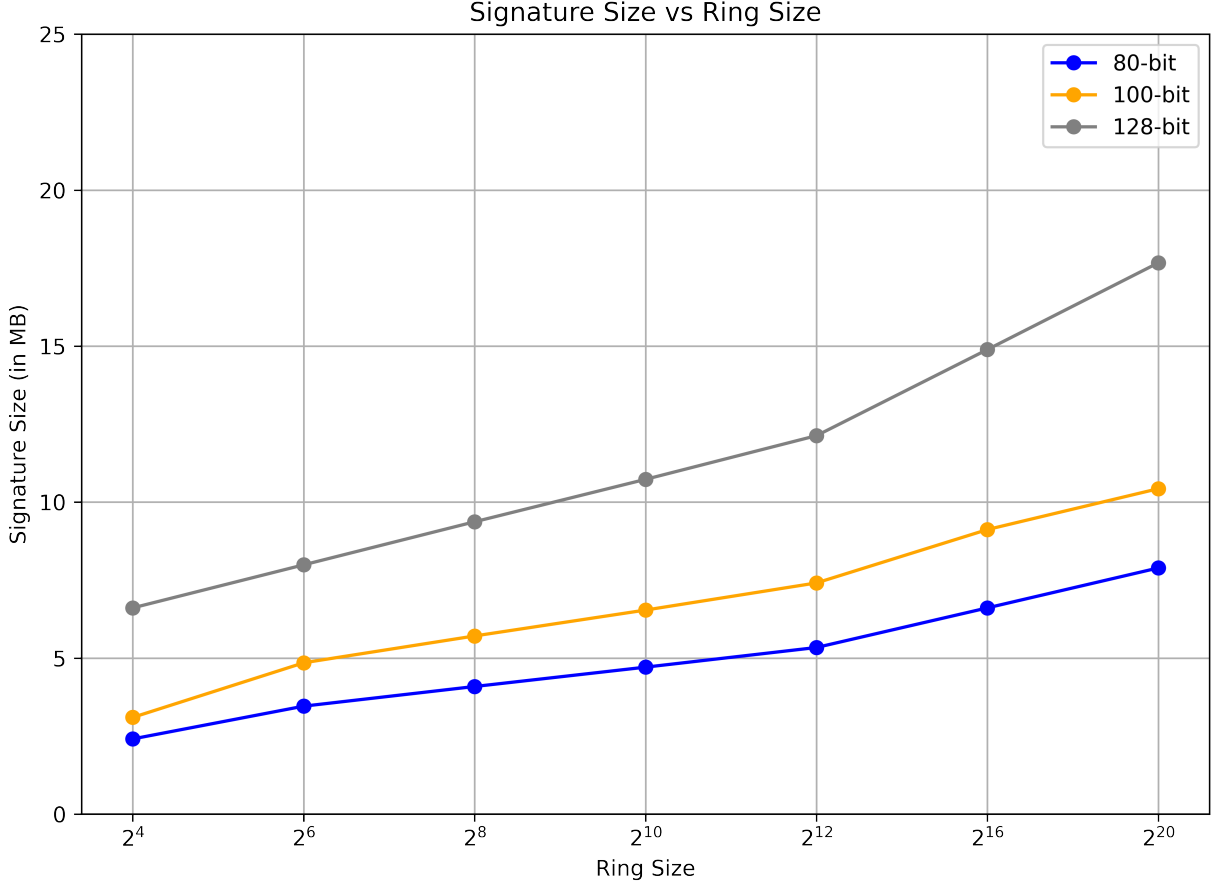


Figure 3: Graphical Representation of Concrete Signature Sizes of LRS^+ Across Security Levels and Ring Sizes

Signature Size analysis with Respect to ENS20 [33]

Esgin et al. [33] proposed an exact zero-knowledge proof system with an optimised proof size by employing algebraic techniques to demonstrate knowledge of a secret vector $\mathbf{x} \in \{-1, 0, 1\}^\eta$ that satisfies the linear relation $\mathbf{A} \cdot \mathbf{x} = \mathbf{z} \bmod q$. We adopt the proof size parameters from the zero-knowledge proof scheme ENS20 (Section 4.2 of [33]) to compute the signature size of our proposed construction, which corresponds to the communication cost of the NIZK proof π . The signature size of our proposed scheme is derived based on specific parameter choices, partially adopted from ENS20 [33]. The details are summarised as follows:

- **Modulus and Dimensions:** We use modulus values $q \approx 2^{32}$, $p \approx 2^{28}$ with dimension parameters $n = 10$, $m_1 = 640$, $m_2 = 783$.
- **Ring Size:** The number of potential users is taken as $N = 2^l = 2^{10}$, where $l = 10$.
- **Secret Vector Length:** The secret vector's length is computed using the formula:

$$\eta = l(2 + 4n + 2nk_1) + m_1 + m_2k = 14507.$$

- **ZKP Parameters (from [33]):** To compute the signature size, we incorporate parameters from [33]:
 - The ring dimension is set to $d = 128$, and the compression parameter $\bar{k} = \frac{128}{\log q} = 4$.
 - For zero-knowledge proof optimization, the bound on $\sigma^i(c)\mathbf{r}_1$ is fixed at $T = 32$.
 - To preserve a root Hermite factor $\delta = 1.0043$, the module ranks are selected as $\kappa' = 9$ for MSIS (Module Short Integer Solution) and $\lambda' = 10$ for MLWE (Module Learning With Errors) (see Section 3.2.4 of [42]).

- The standard deviation of the discrete Gaussian distribution employed for masking is defined as:

$$\mathfrak{s} = 5T \sqrt{\bar{k}(\kappa' + \lambda' + \frac{\eta}{d} + 3)d} = 42220,$$

leading to $\delta_1 = 6\mathfrak{s} \approx 2^{18}$.

- **Signature Size:** Using the above parameters, the signature size for 2^{10} users at the 128-bit security is approximately:

$$\left(\kappa' + \frac{\eta}{d} + 4\right) d * k_1 + \bar{k} \left(\lambda' + \kappa' + \frac{\eta}{d} + 3\right) d \lceil \log 2\delta_1 \rceil \approx 225 \text{ KB}.$$

7 Security analysis of LRS^+

The proposed scheme, LRS^+ , fulfills the fundamental security requirements of a linkable ring signature scheme, including unforgeability, linkable anonymity, linkability, and non-slanderability. The subsequent theorems formally demonstrate the security guarantees provided by LRS^+ .

Theorem 3 *Under the hardness assumption of the Short Integer Solution (SIS), the scheme LRS^+ is considered unforgeable in the random oracle model.*

Proof 2 *We proceed to prove the unforgeability of the linkable ring signature scheme. Let $\mathcal{A}$ be a probabilistic polynomial-time (PPT) adversary. We construct a sequence of games, denoted by $\mathbf{G}_i$, and let P_i represent the probability that $\mathcal{A}$ succeeds in $\mathbf{G}_i$.*

- $\mathbf{G}_0$: This corresponds to the standard unforgeability experiment as defined in 2.4.1.
- $\mathbf{G}_1$: This game is identical to $\mathbf{G}_0$, except that when responding to signing queries, the challenger simulates the zero-knowledge proof π using the simulator of the underlying ZKPoK system. Due to the zero-knowledge property, we have $|P_0 - P_1| \leq \text{neg}(\lambda)$.
- $\mathbf{G}_2$: Same as game $\mathbf{G}_1$, except that when $\mathcal{A}$ produces a forgery $(m^*, \mathbf{R}^*, \sigma^*)$, the challenger attempts to extract a witness $(usk^*, \mathbf{p}^*, ((j_1^*, j_2^*, \dots, j_l^*), \{\mathbf{v}_i^*\}_{i \in [1, l]}, \{\mathbf{w}_i^*\}_{i \in [1, l]}))$ from the proof π^* . If extraction fails, it outputs $\perp$. By the simulation extractability of the proof system, we have $|P_2 - P_1| \leq \text{neg}(\lambda)$.
- $\mathbf{G}_3$: Identical to $\mathbf{G}_2$, except the challenger additionally checks whether $\mathbf{p}^* \in \mathbf{R}^*$ and aborts if so. Due to the accumulator scheme's security, which is based on the hardness of the SIS problem, we get, $|P_3 - P_2| \leq \text{neg}(\lambda)$.
- $\mathbf{G}_4$: Identical to $\mathbf{G}_3$, except that the challenger randomly samples an index $i^* \in \{0, 1, \dots, N - 1\}$ at the beginning and aborts if $\mathbf{p}_{i^*} \neq \text{bin}(\tilde{\mathbf{B}} \cdot usk^* \bmod q)$. Since i^* is independent of $\mathcal{A}$'s choice and $\mathbf{R}^* \subseteq \mathbf{S}_{pk}$, we have $\Pr[\mathbf{p}_{i^*} = \text{bin}(\tilde{\mathbf{B}} \cdot usk^*)] = 1/N$. Therefore, we obtain $P_4 = P_3/N$.

If the probability P_4 is non-negligible, then an efficient adversary $\mathcal{B}$ can be constructed to break the soundness of the underlying ZKAoK by extracting a valid solution from the forgery. Hence, P_4 must be negligible. From the above games, we have

$$|NP_4 - P_0| \leq \text{neg}(\lambda),$$

which implies that P_0 is negligible, thereby completing the proof.

Theorem 4 *Under the hardness assumption of the Learning With Rounding (LWR) problem, the scheme LRS^+ achieves linkable-anonymity in the random oracle model.*

Proof 3 *We now give proof that the linkable ring signature scheme ensures linkable anonymity. Let P_i denote the probability that the adversary $\mathcal{A}$ succeeds in game $\mathbf{G}_i$.*

- $\mathbf{G}_0$: This is the actual linkable-anonymity experiment as in 2.4.1.
- $\mathbf{G}_1$: Identical to $\mathbf{G}_0$, except the challenger simulates the proof π^* in the challenge signature σ^* using the zero-knowledge property of the underlying ZKAoK system. Hence,

$$|P_1 - P_0| \leq \text{neg}(\lambda).$$

- $\mathbf{G}_2$: Same as $\mathbf{G}_1$, but the challenger selects two challenge indices $i_0^*, i_1^* \xleftarrow{\$} [0, N-1]$ uniformly at random. If either $i_0^* \in Q$ or $i_1^* \in Q$, where Q is the set of queried indices (via signing or corruption), the challenger the experiment and outputs a random bit $b' \in \{0, 1\}$. Otherwise, it proceeds as in $\mathbf{G}_1$.

The probability that both i_0^* and i_1^* are unqueried is:

$$\Pr[i_0^*, i_1^* \notin Q] = \left(1 - \frac{|Q|}{n}\right)^2$$

Hence, the success probability of the adversary in this game is:

$$P_2 = \left(1 - \frac{|Q|}{N}\right)^2 \cdot P_1 + \left(1 - \left(1 - \frac{|Q|}{N}\right)^2\right) \cdot \frac{1}{2}$$

where P_1 is the success probability in $\mathbf{G}_1$. The second term accounts for the abort case, where the adversary wins with probability $\frac{1}{2}$ due to receiving no useful information. For the worst-case upper bound, we take $|Q| = N - 1$, we get

$$P_2 = \frac{1}{N^2} \cdot P_1 + \left(1 - \frac{1}{N^2}\right) \cdot \frac{1}{2} = \frac{1}{2} + \frac{1}{N^2} \left(P_1 - \frac{1}{2}\right).$$

- $\mathbf{G}_3$: Same as $\mathbf{G}_2$, but the public key $\mathbf{p}_{i_0}$ is sampled uniformly at random, and $\tilde{t}^*$ is sampled randomly from the wPRF range if $b = 0$. The $\mathbf{G}_2$ and $\mathbf{G}_3$ are indistinguishable due to the weak pseudorandomness function, and we obtain

$$|P_3 - P_2| \leq \text{neg}(\lambda).$$

- $\mathbf{G}_4$: Identical to $\mathbf{G}_3$, except that $\mathbf{p}_{i_1^*}$ is randomly selected and $\tilde{t}^*$ is sampled randomly when $b = 1$ from range of wPRF. Again, due to the weak pseudorandomness function ensures

$$|P_4 - P_3| \leq \text{neg}(\lambda).$$

In $\mathbf{G}_4$, $\mathcal{A}$'s view is independent of b , so $P_4 = \frac{1}{2}$. Thus,

$$|P_0 - \frac{1}{2}| \leq \text{neg}(\lambda),$$

which concludes the proof of linkable anonymity.

Theorem 5 Assuming the hardness of the Learning With Rounding (LWR) problem, the scheme LRS^+ attains linkability within the random oracle model.

Proof 4 We now prove the linkability of our LRS^+ scheme by defining the following sequence of games. Let P_i denote the probability that the adversary $\mathcal{A}$ wins in $\mathbf{G}_i$.

- $\mathbf{G}_0$: This game corresponds to the actual linkability experiment as specified in Section 2.4.1.
- $\mathbf{G}_1$: This game is identical to $\mathbf{G}_0$ except that, during responses to the signing oracle, the challenger uses the simulator to generate the proof π included in the signature σ . By leveraging the security of the underlying ZKAoK protocols, we obtain the following bound:

$$|P_1 - P_0| \leq \text{neg}(\lambda).$$

- $\mathbf{G}_2$: Same as $\mathbf{G}_1$, but now when $\mathcal{A}$ succeeds, the challenger tries to extract valid witnesses from the submitted signatures. Specifically, it extracts $(\text{usk}_1^*, \mathbf{p}_1^*, ((j_{1,1}^*, j_{1,2}^*, \dots, j_{1,l}^*), \{\mathbf{v}_{1,i}^*\}_{i \in [1,l]}, \{\mathbf{w}_{1,i}^*\}_{i \in [1,l]}))$ from σ_1^* and $(\text{usk}_2^*, \mathbf{p}_2^*, ((j_{2,1}^*, j_{2,2}^*, \dots, j_{2,l}^*), \{\mathbf{v}_{2,i}^*\}_{i \in [1,l]}, \{\mathbf{w}_{2,i}^*\}_{i \in [1,l]}))$ from σ_2^* . If either extraction fails or the witnesses are invalid, the challenger outputs 0 (adversary fails). Due to the simulation extractability of the ZKPoK systems, with all but negligible probability, the challenger extracts valid witnesses. Also, the queries $(\mathbf{m}_1^*, \mathbf{R}_1^*, \sigma_1^*)$ and $(\mathbf{m}_2^*, \mathbf{R}_2^*, \sigma_2^*)$ would have appeared before in signing the oracle with negligible probability. Thus,

$$|P_2 - P_1| \leq \text{neg}(\lambda).$$

- $\mathbf{G}_3$: This game follows $\mathbf{G}_2$ with an additional verification step: when the adversary $\mathcal{A}$ succeeds, the challenger also verifies whether $\mathbf{p}_1^* \in \mathbf{R}_1^*$ and $\mathbf{p}_2^* \in \mathbf{R}_2^*$. If either check fails, the challenger outputs 0. The difference between $\mathbf{G}_3$ and $\mathbf{G}_2$ is negligible, relying on the security of the accumulator and the hardness of the SIS problem. Formally,

$$|P_3 - P_2| \leq \text{neg}(\lambda).$$

- $\mathbf{G}_4$: Identical as $\mathbf{G}_3$, but the challenger now checks if $\tilde{t}_1^* = \tilde{t}_2^*$, then outputs 0. The $\mathbf{G}_3$ and $\mathbf{G}_4$ are identical unless $\mathcal{A}$ wins $\mathbf{G}_3$ with $\text{usk}_1^* = \text{usk}_2^*$. But this is not possible due to the uniqueness of wPRF. Therefore,

$$|P_4 - P_3| \leq \text{neg}(\lambda).$$

- $\mathbf{G}_5$: Same as $\mathbf{G}_4$, except that the challenger samples a random index $i^* \in \{0, 1, \dots, N-1\}$ at the beginning and aborts if $\mathbf{p}_{i^*} \neq \text{bin}(\tilde{\mathbf{B}} \cdot \text{usk}^* \bmod q)$. Since i^* is chosen independently of $\mathcal{A}$'s view, $\mathbf{R}^* \subseteq \mathbf{S}_{pk}$ and $\Pr[\mathbf{p}_{i^*} = \text{bin}(\tilde{\mathbf{B}} \cdot \text{usk}^*)] = 1/N$. Thus, we have $P_5 = \frac{P_4}{N}$.

If the probability P_5 is non-negligible, then we could build an efficient adversary $\mathcal{B}$ that breaks the soundness of the underlying ZKAoK by extracting a valid solution from the forgery. Hence, P_5 must be negligible. From the above games, we have

$$|NP_4 - P_0| \leq \text{neg}(\lambda),$$

which implies that P_0 is negligible. This concludes the proof.

Theorem 6 Under the hardness assumption of the Learning With Rounding (LWR) problem, the scheme LRS^+ achieves non-slanderability in the random oracle model.

Proof 5 We proceed to prove the non-slanderability of our linkable ring signature scheme by defining a sequence of games, where P_i denotes the probability that adversary $\mathcal{A}$ succeeds in $\mathbf{G}_i$.

- $\mathbf{G}_0$: This is the actual non-slanderability experiment as defined in 2.4.1.
- $\mathbf{G}_1$: Identical to $\mathbf{G}_0$, except that the challenger samples an index $i^* \leftarrow \{0, 1, \dots, N-1\}$ at the beginning and aborts if $i^* \neq j$. Since i^* is chosen independently of $\mathcal{A}$'s view, we have $\Pr[i^* = j] = \frac{1}{N}$. Therefore, $P_1 = \frac{P_0}{N}$.
- $\mathbf{G}_2$: Similar to $\mathbf{G}_1$, except that during signature verification, the challenger obtains the proof π in the signature σ using a simulator. The indistinguishability between $\mathbf{G}_1$ and $\mathbf{G}_2$ follows from the zero-knowledge property of the underlying ZKAoK system. Thus,

$$|P_2 - P_1| \leq \text{neg}(\lambda).$$

- $\mathbf{G}_3$: Same as $\mathbf{G}_2$, but when $\mathcal{A}$ succeeds, the challenger tries to extract the witness $(\text{usk}^*, \mathbf{p}^*, ((j_1^*, j_2^*, \dots, j_l^*), \{\mathbf{v}_i^*\}_{i \in [1, l]}, \{\mathbf{w}_i^*\}_{i \in [1, l]}))$ from the proof π^* in the forged signature σ^* . If the extraction fails, the challenger outputs 0 (i.e., $\mathcal{A}$ fails). A witness is considered valid if it satisfies the relation defined in the signing algorithm. Note that in our scheme, the hash of the message m (along with other elements) is part of the statement in π , so with negligible probability, a different π will satisfy the same relation. Moreover, with overwhelming probability, the pair $(\mathbf{R}^*, \tilde{t}^*)$ extracted from the tuple $(i^*, m^*, \mathbf{R}^*, \sigma^*)$, which was not submitted to the signing oracle, does not correspond to any previously simulated proof. Due to the simulation extractability property of the underlying ZKP system, the challenger can extract a valid witness with only negligible failure. Hence, we have

$$|P_3 - P_2| \leq \text{neg}(\lambda).$$

In $\mathbf{G}_3$, we have $\tilde{t} = \text{bin}(\mathcal{F}_{\text{usk}}(\tilde{\mathbf{T}}) \bmod p)$ if $\mathcal{A}$ wins, where $\tilde{t}$ is the unique tag for σ . Thus, by the property of wPRF, P_3 is negligible.

This concludes the proof of non-slanderability.

8 Conclusion

Our proposed linkable ring signature scheme, grounded in exact zero-knowledge proofs, achieves efficient signature sizes and demonstrates strong potential for real-world deployment. However, there remains considerable room for

enhancement. By using more advanced proof systems and adjusting some parameters, we can make the scheme even more compact and useful. Adding features like revocation through accumulators, time-based tags to prevent repeated reports, and combining linkability with threshold signatures can help to solve many real-life privacy and accountability problems. These ideas offer promising directions for future work.

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